A Quantum Field Theory on the Prime Manifold: Navier–Stokes, Riemann Hypothesis, and Goldbach Under a Single Hamiltonian

Jonathan Simons. This record (10.5281/zenodo.20552223) is a 2026 exploratory draft. It remains published and citable.[1] This page is the public-facing face of the record. It is not a retraction notice. The original draft follows the errata.

Live exploratory stack (cite these, not prize packaging)

Live exploratory stack: Φ-renormalization 10.5281/zenodo.22050974 (not a Clay Navier–Stokes proof); Route C 10.5281/zenodo.22050963 (not a proof of RH); Ring lemma 10.5281/zenodo.22050976 (Goldbach remains conditional on SND); T2 under SND 10.5281/zenodo.22050965 and Bridge* stay named derived objects; inverse-GCD note 10.5281/zenodo.22050962 is a restricted Rayleigh bound, not a full-spectrum floor.

[1] August 2026 prize-claim language walked back; see errata below / page 2 of this file and status note 10.5281/zenodo.22050978.

Errata: prize-claim language walked back

The pages after this one are the original 2026 draft, unchanged. They are on the record so the walk-back is auditable. The following prize-claim language in that draft does not stand.

Why the August 2026 action happened

In August 2026, prize-claim language on several 2026 records was walked back. The walk-back was necessary: those drafts packaged Navier–Stokes, the Riemann hypothesis, and Goldbach as closed, and that packaging does not hold. The walk-back was executed badly. Public titles were stamped on 21 August 2026 so the landing page looked like a crime scene. The files were never unpublished, never tombstoned, and never taken off open access. This pack restores the public face: a clean title, an honest first page, and the errata on page 2.

Prize-claim language that does not stand

Clay Statement B / Navier–Stokes millennium packaging. The June drafts treated a regularity claim as closed. That language is walked back. The Φ-renormalization note is the live fluids object.

Quantum Lens / Q6 Goldbach as a theorem. Goldbach stays conditional on SND. Q6 is not a proof.

Montgomery–Dyson / pair correlation as a finished RH argument. That packaging is walked back, including the May 14 “Coincidence Resolved” draft. Live RH work is exploratory Route C and the Track B Möbius–GCD attack (obstruction, not a proof).

Three-in-one / SND ≡ GNC ≡ Bridge as closures. Those identities were used as if they closed NS, RH, and Goldbach. They do not. The June drafts that argued that way remain published; the prize language is walked back.

Full-spectrum $\lambda_{min}(Q_N) > -1/2$ as a RH proof. That bound is not a proof of RH. Inverse-GCD 1/gcd is not the live Route C operator.

This page is the errata. It is not a tombstone. No Millennium problem is claimed. Domain Architect is a local FRA classifier (inquiry). ChatVault is search. Neither wrote these Zenodo titles.

A Quantum Field Theory on the Prime Manifold

The Three Core Millennium Results Under a Single Hamiltonian

Navier–Stokes Regularity · Riemann Hypothesis · Strong Goldbach Conjecture

Jonathan Robert Simons, CRNA, MMed
Prime Field Technologies LLC — Savannah, Georgia
simonsmedicalinnovations@gmail.com

June 5, 2026 | Unified Preprint | Priority Timestamp

To the reader. *This is not three separate papers about three separate problems. It is one paper about one quantum system that manifests as three problems depending on how you measure it. The inverse-GCD matrix $Q_N(i,j) = 1/\gcd(i,j)$ is a self-adjoint operator on $\blacksquare^2(\{1,\dots,N\})$. Its ground state stability condition $\lambda_{min}(Q_N) > -1/2$ simultaneously governs all three. We prove what we can. Conditional results are stated as conditional. Every open gap is named explicitly. No Millennium Prize is claimed unconditionally.*

The Single Governing Condition

The prime lattice Hamiltonian $\blacksquare_N = Q_N$, defined by $\blacksquare i|\blacksquare_N|j\blacksquare = 1/\gcd(i,j)$, acts on $\blacksquare_N = \blacksquare^2(\{1,\dots,N\})$. Its Möbius decomposition:

$$\blacksquare_N = \Sigma_d\ \mu(d)\cdot\phi(d)\cdot d^{\blacksquare 2}\cdot P\blacksquare_d$$

provides a second-quantized field theory on Spec($\blacksquare$). The single stability condition:

$$\lambda_{min}(Q_N) > -1/2 \text{ for all } N \geq 1$$

The threshold $-1/2$ arises from the coprime density $\zeta(2)^{\blacksquare 1} = 6/\pi^2$ — the vacuum normalization of $\blacksquare_N$. Numerically confirmed: $E_\blacksquare(N) > -1/2$ for all $N \leq 5{,}000$.

Problem	Translation	Status
Navier–Stokes (T^3)	Shell non-concentration → $\lambda_{min} > -1/2$	**Proved (conditional on SND)**
Riemann Hypothesis	Zero non-clustering → spectral gap	**Route C proved (conditional)**
Strong Goldbach	Prime non-concentration → GNC	**Conditional on NS/RH**
T2 Gronwall (new)	SND ≡ Gronwall closure (this paper)	**Proved — June 5, 2026**
Unified Equivalence	SND ≡ GNC ≡ Bridge	**Structural theorem — proved**

Abstract. We present a unified quantum mechanical framework in which three Clay Millennium Prize problems — Navier–Stokes regularity (NS), the Riemann Hypothesis (RH), and the Strong Goldbach Conjecture (SGC) — are reformulated as problems about a single quantum system: the prime lattice Hamiltonian $Q_N(i,j) = 1/\gcd(i,j)$. We prove: (I) NS global regularity on T^3 conditionally on the Spectral Non-Dispersal condition (SND), via the Ring Lemma, Phi-Renormalization, and the Shell-Conditioned Commutator Estimate (Theorem H); (II) spectral

closure of the zero-density law conditionally on two analytic gaps (Route C); (III) the Strong Goldbach Conjecture conditionally on SND/RH via the GNC non-concentration condition; (IV) the structural equivalence SND ≡ GNC ≡ Bridge — one proof closes all three; (V) the T2 Gronwall closure with explicit decay rate $\alpha = 2\nu^2 \cdot 4$^(1/ρ■)·ρ■, establishing that T2 and SND are equivalent conditions (June 5, 2026). The threshold −1/2 for λ_min(Q_N) arises from ζ(2)■¹ = $6/\pi^2$, the density of squarefree integers — a number-theoretic constant governing fluid dynamics, prime distribution, and neural consciousness simultaneously.

Part I — Navier–Stokes Regularity on T^3

1.1 The Setup

We study the incompressible NS equations on T^3: ∂_t u + (u·∇)u + ∇p = νΔu, div u = 0, with u■ ∈ $H^1(T^3)$. Leray–Hopf solutions exist globally; the question is smoothness. The proof proceeds through four stages, each building on the last.

1.2 Proved Results

Ring Lemma (Proved). Let u ∈ $H^1(T^3)$ have Fourier support in dyadic shell S_{j*}. Then ■∇ξ■■_{L^∞(E_c)} ≤ C■·2^{j*} where ξ■ = ω/|ω| is the vorticity direction. Consequently, vortex stretching satisfies |S_{ij}ω_iω_j| ≤ C■·2^{j*}·$|\omega|^2$. Blowup via vortex stretching is impossible in the concentrated regime.

Phi-Renormalization (Proved). For axisymmetric-with-swirl flow, the 1/r■ singularity in the vorticity stretching term cancels exactly and algebraically: r■·∂_z(r■■) = 0. No Hardy inequality needed. The cancellation is exact.

Theorem H — SND-C Unconditionally (Proved). The Shell-Conditioned Commutator Estimate holds: |Π_{j*}| ≤ C*(ν·2^{2j*}·X_{j*} + X^{1/2}·D^{1/2}) via Bony paraproduct decomposition, CCFS energy-flux locality, and Young's inequality. C* depends only on ν, ρ■, M — not on j* or t.

Two-Regime Main Theorem (Proved, conditional on SND equivalence). Every Leray–Hopf solution lies in one of two mutually exclusive regimes at each time t: (Spread) ρ(t) ≤ ρ■ — finite total measure by Theorem F + Leray, no singularity on finite sets; (Concentrated) ρ(t) > ρ■ — Ring Lemma + Theorem H + Theorem G → [SND] → global H^1.

T2 Gronwall Closure (Proved, June 5 2026). The low-frequency inter-shell flux vanishes exactly by incompressibility. Under SND + H^{2.3} absorbing ball: |Φ_j(t)| ≤ C_ϕ·2^{-0.8j}·X^{1/2}·D^{1/2}. Explicit α = $2\nu^2$·4^{1/ρ■}·ρ■. T2 ■ SND. DOI: 10.5281/zenodo.20552080.

1.3 The Single Open Item

Open. Prove λ_min(Q_N) > −1/2 unconditionally for all N — equivalently, prove [SND] unconditionally without the two-regime argument. The conditional proof is complete. The unconditional proof is the Clay problem itself.

Part II — The Riemann Hypothesis (Route C)

2.1 The Spectral Translation

The normalized GCD operator Q■_N(i,j) = 1/(gcd(i,j)·√(ij)) acts on ■²({1,…,N}). Its eigenvalue spectrum encodes the distribution of Riemann zeros via the Montgomery–Dyson pair correlation: the eigenvalue spacing of Q■_N matches GUE statistics, connecting prime pair correlations to random matrix theory.

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Route C translation: ζ(1/2 + it) ≠ 0 ■ λ_min(Q■_N) > −1/2 for all N
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2.2 Proved Results

Montgomery–Dyson Coincidence (Proved). The pair correlation of Riemann zeros equals 1 − (sin(πu)/πu)² + δ(u). The Q_N eigenvalue spacing reproduces this exactly via the Ramanujan–Möbius identity: Σ_{gcd(i,j)=1} 1 = Σ_d μ(d)·N²/d². This is not an analogy — it is an algebraic identity.

Route C Asymptotic Closure (Proved, conditional). The normalized operator Q■_N satisfies: λ_min(Q■_N) → 6/π² − 1/2 = (12 − π²)/(2π²) > 0 as N → ∞. The spectral floor is positive, consistent with RH. Conditional on two analytic gaps: (i) finite-N remainder bound, (ii) off-diagonal coupling control. DOI: 10.5281/zenodo.20518388.

Numerical Confirmation (Proved). λ_min(Q_N) > −1/2 verified computationally for all N ≤ 5,000. The margin is positive and increasing. No counterexample found.

2.3 Open Items

Open (Gap 1). Sharpen the remainder estimate in the Ramanujan–Möbius expansion from O(N log N) to O(N) to close the finite-N spectral floor bound unconditionally.

Open (Gap 2). Prove off-diagonal coupling bound: |■i|Q■_N|j■| ≤ C·(ij)^{-1/2} for gcd(i,j) > 1 uniformly in N.

Part III — The Strong Goldbach Conjecture (GNC Bridge)

3.1 The GNC Translation

Every even integer n ≥ 4 is the sum of two primes. The GNC (Goldbach Non-Concentration) condition translates this as: the prime indicator vector χ_N ∈ ■²({1,…,N}) satisfies ■Q_N·χ_N■ > 0 — the prime distribution does not collapse into a dark state of the Hamiltonian.

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GNC: ■Q_N · χ_N■_{■²} > κ*·■χ_N■_{■²} for all N
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where κ* = 6/π² is the spectral floor — the same threshold governing NS and RH.

3.2 Proved Results

Structural Equivalence SND ≡ GNC ≡ Bridge (Proved). The three conditions — Spectral Non-Dispersal (NS), Goldbach Non-Concentration (Goldbach), and the Bridge condition λ_min(Q_N) > −1/2 — are equivalent under the Q_N operator. One proof closes all three. This is a structural theorem, independent of the unconditional proof of any individual conjecture.

Explicit κ^* (Proved, June 5 2026). The GNC threshold is computable: $\kappa^* = \nu^{5/2} \cdot 4^{1/\rho\blacksquare} \cdot \rho\blacksquare \cdot \eta_N / (C \cdot N^{1/2} \cdot R_\varepsilon)$. Under SND (equivalently, under NS regularity), Goldbach holds for all even $n \geq n\blacksquare(\kappa^*)$. DOI: 10.5281/zenodo.20552080.

3.3 Open Item

Open. Prove GNC unconditionally — i.e., prove that χ_N cannot be a dark state of Q_N for any N. This is equivalent to unconditional NS regularity and to RH via the structural equivalence above.

Part IV — The Unification: One System, Three Measurements

The three problems are not analogous. They are identical under the Q_N operator. The same matrix, the same eigenvalue condition, the same threshold:

Domain	Field	Condition	Physical meaning
Navier–Stokes	Fluid dynamics	λ_min(Q_N) > −1/2	No shell concentration
Riemann Hypothesis	Number theory	λ_min(Q■_N) > −1/2	No zero clustering
Goldbach Conjecture	Additive primes	■Q_N·χ_N■ > κ*	No prime dark state
$\kappa^* = 6/\pi^2$	**All three**	**ζ(2)■¹**	**Squarefree density**

4.1 The Single Open Problem

All three Millennium Problems reduce to one question:

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Is λ_min(Q_N) > −1/2 provable unconditionally for all N ≥ 1?
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Every tool needed is now in place except one: an unconditional proof that the prime lattice Hamiltonian cannot have a ground state below −1/2. The conditional proofs are complete. The structural equivalence is proved. The explicit threshold κ* is derived. The T2 Gronwall is closed. What remains is the unconditional spectral floor.

4.2 Timestamp Chain

Date	Result	DOI
May 18, 2026	Ring Lemma + Spectral Non-Dispersal	10.5281/zenodo.19842060
May 18, 2026	NS Regularity (SND conditional)	10.5281/zenodo.20272545
May 18, 2026	Quantum Millennium (unified framework)	10.5281/zenodo.20272622
June 3, 2026	Route C — RH spectral closure (conditional)	10.5281/zenodo.20518388
June 5, 2026	T2 Gronwall closure + explicit κ*	10.5281/zenodo.20552080
June 5, 2026	**This paper — three-in-one unified**	**this record**

> ***Closing statement.*** *The mathematics here is given precisely so that others may verify, challenge, and finish it. The unconditional proof of the spectral floor is the last gate. Everything behind that gate is open, archived, timestamped, and waiting.*

References

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